

Proof Using Resolution

Outline

I. Rule of resolution

II. Conjunctive normal forms

III. Resolution refutation

I. Resolution

An inference algorithm i is

sound if $KB \models \alpha$ whenever $KB \vdash_i \alpha$

complete if $KB \vdash_i \alpha$ whenever $KB \models \alpha$

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- ◆ Inference rules covered so far are sound.
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|

single inference rule

Wumpus World Revisited

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
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KB:

$$R_1: \neg P_{1,1}$$

$$R_2: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

$$R_3: B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$$

$$R_4: \neg B_{1,1}$$

$$R_5: B_{2,1}$$

Rules

$$R_2: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

biconditional elimination

$$R_6: (B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$$

and-elimination

$$R_7: (P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1}$$

logical equivalence

$$R_4: \neg B_{1,1}$$

$$R_8: \neg B_{1,1} \Rightarrow \neg(P_{1,2} \vee P_{2,1})$$

modus ponens

$$R_9: \neg(P_{1,2} \vee P_{2,1})$$

De Morgan's rule

$$R_{10}: \neg P_{1,2} \wedge \neg P_{2,1}$$

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Added to KB via inferences

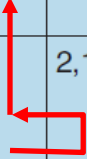
(cont'd)

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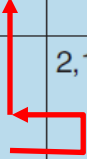
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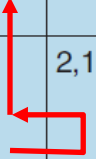
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[1,1] → [1,2]: stench but no breeze

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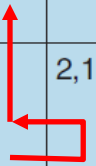
Add to KB:

$$R_{11}: \neg B_{1,2}$$

$$R_{12}: B_{1,2} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{1,3})$$

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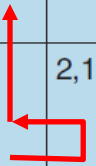
$$\neg B_{1,2} \Rightarrow \neg P_{1,1} \wedge \neg P_{2,2} \wedge \neg P_{1,3}$$

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$$R_3: B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$$



biconditional elimination

$$R_5: B_{2,1}$$

$$R_{15}: P_{1,1} \vee P_{2,2} \vee P_{3,1}$$

Resolvent

$$R_{13}: \neg P_{2,2}$$

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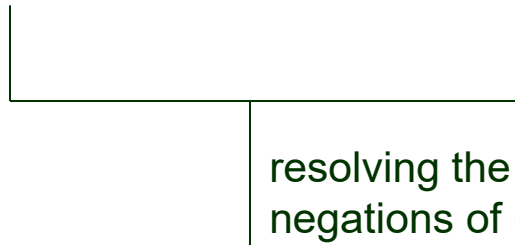


resolving the two literals that are
negations of each other

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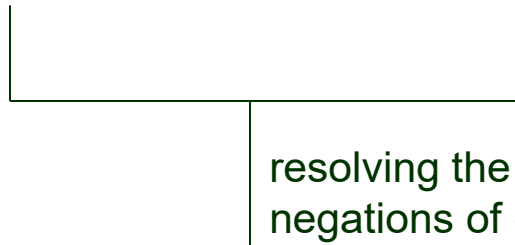
resolving the two literals that are
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$R_{16}: P_{1,1} \vee P_{3,1}$ (*resolvent*)

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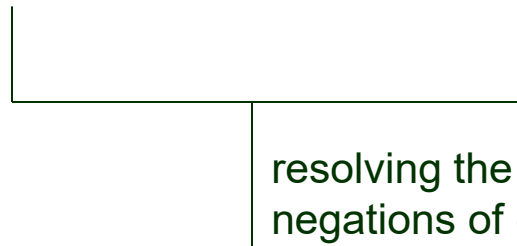
$R_{16}: P_{1,1} \vee P_{3,1}$ (*resolvent*)

If there's a pit in one of [1,1], [2,2], and [3,1] and it's not in [2,2], then it's in [1,1] or [3,1].

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$R_{16}: P_{1,1} \vee P_{3,1}$



$R_{17}: P_{3,1}$

Simple Resolution Rule

$$\frac{l_1 \vee \cdots \vee l_i \vee \cdots \vee l_k, \quad m}{l_1 \vee \cdots \vee l_{i-1} \vee l_{i+1} \vee \cdots \vee l_k}$$

(l_i and m are complementary literals, i.e., $l_i = \neg m$ or $m = \neg l_i$.)

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Since m is true, then l_i must be false. But one of $l_1, \dots, l_k$ must be true. Therefore, we can exclude l_i and assert that one of the remaining $k - 1$ literals must be true.

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l_i and m_j are complementary literals:

$$l_1 \vee \cdots \vee l_i \vee \cdots \vee l_k, \quad m_1 \vee \cdots \vee m_j \vee \cdots \vee m_n$$

$$l_1 \vee \cdots \vee l_{i-1} \vee l_{i+1} \vee \cdots \vee l_k \vee m_1 \vee \cdots \vee m_{j-1} \vee m_{j+1} \vee \cdots \vee m_n$$

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If l_i is true, then m_j is false. Hence $m_1 \vee \cdots \vee m_{j-1} \vee m_{j+1} \vee \cdots \vee m_n$ must be true.
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$$\frac{P_{1,1} \vee P_{3,1}, \quad \neg P_{1,1} \vee \neg P_{2,2}}{P_{3,1} \vee \neg P_{2,2}}$$

One Pair at a Time

Only one pair of complementary literals can be resolved at each step.

$$P \vee \neg Q \vee R, \quad \neg P \vee Q$$

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Incorrect conclusion!

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II. Conjunctive Normal Form

The resolution rule applies to clauses only.

Conjunctive normal form (CNF): a conjunction of clauses

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$$CNFSentence \rightarrow Clause_1 \wedge \dots \wedge Clause_n$$

$$Clause \rightarrow Literal_1 \vee \dots \vee Literal_m$$

$$Fact \rightarrow Symbol$$

$$Literal \rightarrow Symbol \mid \neg Symbol$$

$$Symbol \rightarrow P \mid Q \mid R \mid \dots$$

II. Conjunctive Normal Form

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$$Fact \rightarrow Symbol$$

$$Literal \rightarrow Symbol \mid \neg Symbol$$

$$Symbol \rightarrow P \mid Q \mid R \mid \dots$$

Every sentence of propositional logic is equivalent to a CNF.

Converting to CNF

1. Eliminate $\Leftrightarrow$.

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$$\begin{array}{c} B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}) \\ \downarrow \\ (B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1}) \end{array}$$

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Converting to CNF

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$$\begin{array}{l} \neg(\neg \alpha) \equiv \alpha \\ \neg(\alpha \vee \beta) \equiv (\neg \alpha \wedge \neg \beta) \\ \neg(\alpha \wedge \beta) \equiv (\neg \alpha \vee \neg \beta) \end{array}$$

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CNF Conversion Algorithm (Optional)

- ◆ Parse every propositional sentence in the KB as an arithmetic expression to construct an expression tree (COMS 2280).

Operators (connectives): $\neg$, $\wedge$, $\vee$, $\Rightarrow$, $\Leftrightarrow$

Operands (atomic sentences): $P, Q, R, S, T, \dots$

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(infix expression)

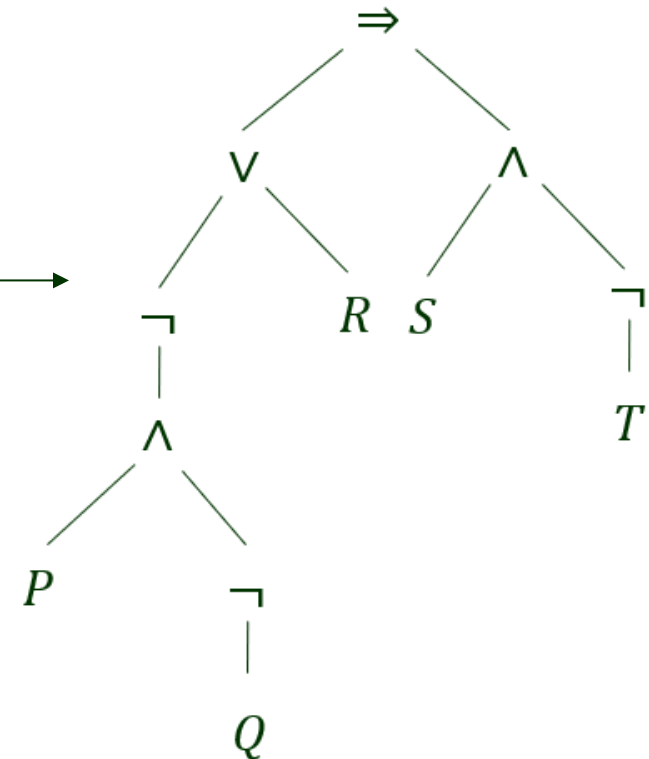
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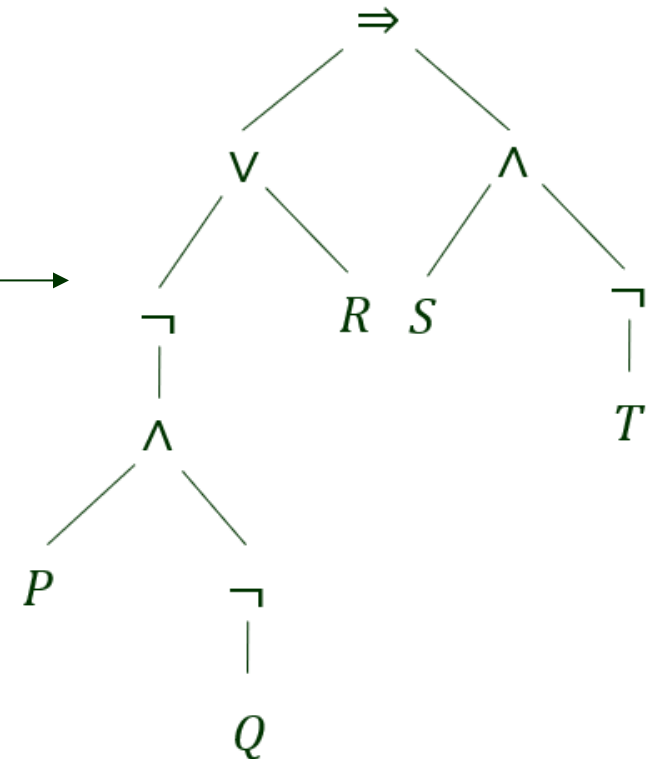
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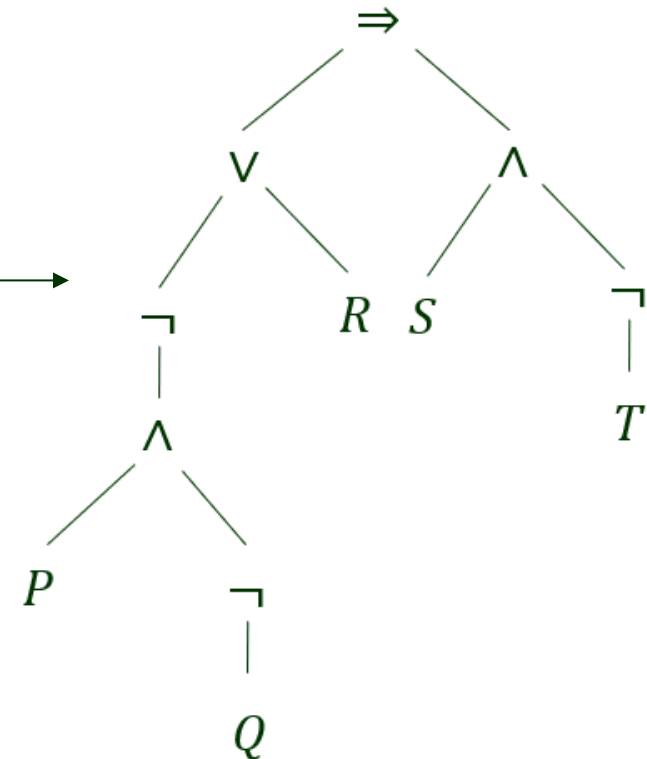
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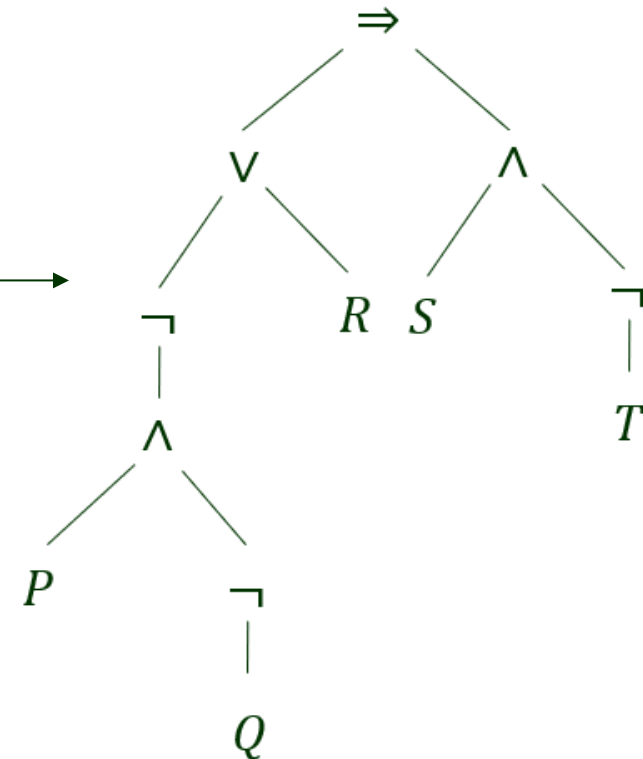
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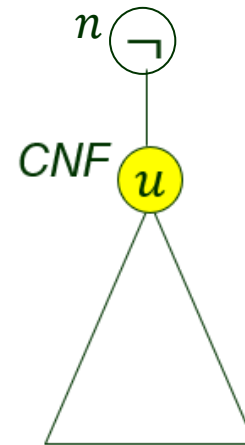
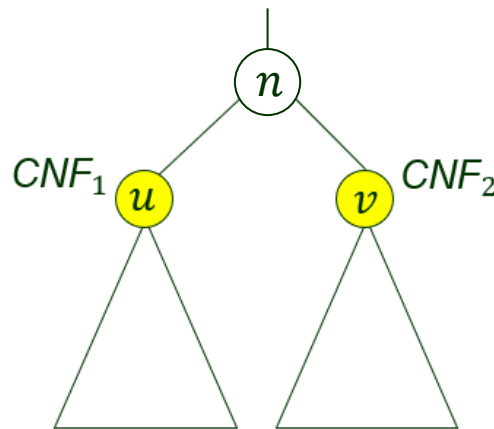
$PQ\neg\wedge\neg R\vee ST\neg\wedge\Rightarrow$
(postfix expression)



- Instead of outputting an operator after its two operands in the postfix format, now you just make the logical operator the parent of the two roots of the subtrees that store the same operator's subexpression operands.

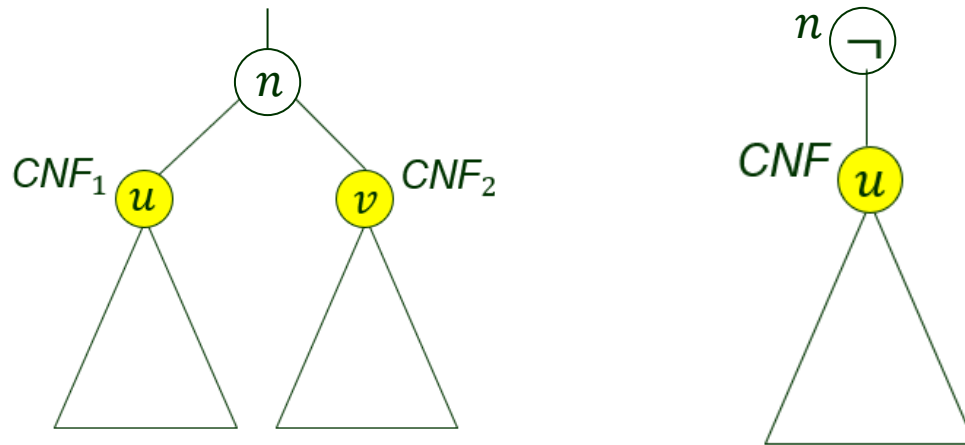
Postorder Traversal

- ◆ Perform a postorder traversal of the expression tree.
 - When visiting an internal node n (representing a connective), its left and right children (or its unique child in the case of a $\neg$ node) store the CNFs for the expressions represented by the left and right subtrees.



Postorder Traversal

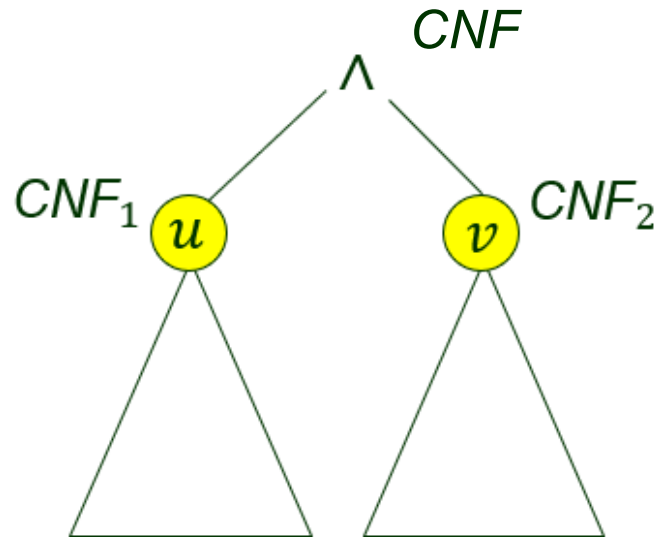
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 - When visiting an internal node n (representing a connective), its left and right children (or its unique child in the case of a $\neg$ node) store the CNFs for the expressions represented by the left and right subtrees.



- Conversion is done in five cases depending on the logical operator stored at n .

Case 1

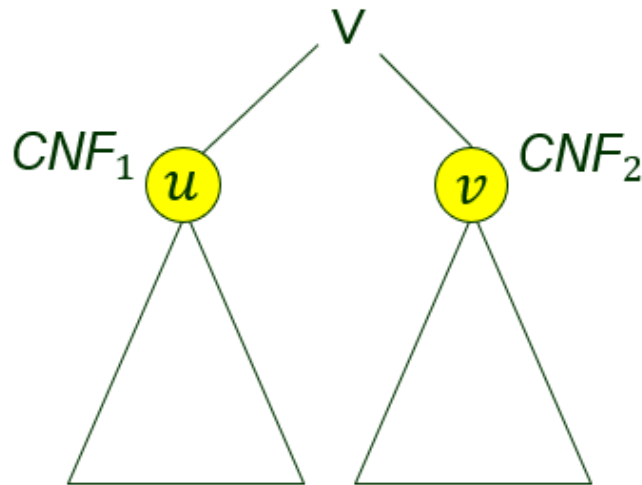
The node n stores Λ .



$$CNF \equiv CNF_1 \wedge CNF_2$$

Case 2

The node n stores V .



$$CNF_1 \equiv C_1 \wedge \cdots \wedge C_k$$

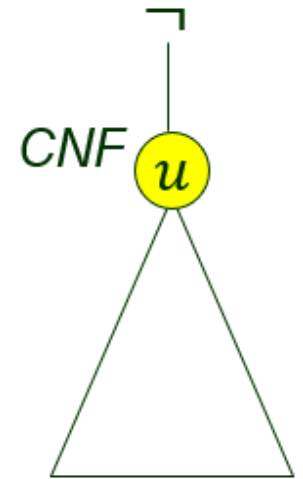
$$CNF_2 \equiv C'_1 \wedge \cdots \wedge C'_m$$

$$CNF \equiv CNF_1 \vee CNF_2 \equiv \bigwedge_{i=1, \dots, k} \bigvee_{j=1, \dots, m} (C_i \vee C'_j)$$

Case 3

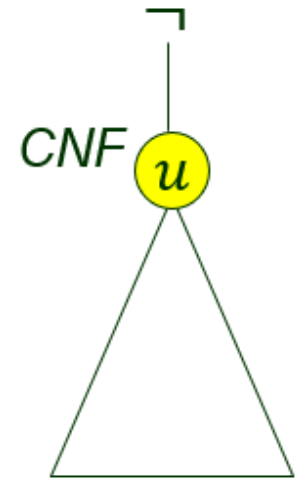
The node n stores $\neg$.

$$CNF \equiv (l_{11} \vee \cdots \vee l_{1k_1}) \wedge (l_{21} \vee \cdots \vee l_{2k_2}) \wedge \cdots \wedge (l_{r1} \vee \cdots \vee l_{rk_r})$$



Case 3

The node n stores $\neg$.



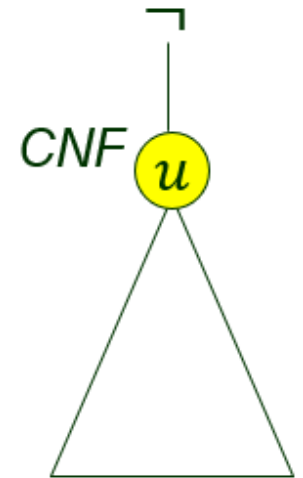
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$$\neg CNF \equiv \bigwedge_{\substack{1 \leq j_1 \leq k_1 \\ \vdots \\ 1 \leq j_r \leq k_r}} (\neg l_{1j_1} \vee \neg l_{2j_2} \vee \cdots \vee \neg l_{rj_r})$$

Case 3

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$$CNF \equiv (l_{11} \vee \cdots \vee l_{1k_1}) \wedge (l_{21} \vee \cdots \vee l_{2k_2}) \wedge \cdots \wedge (l_{r1} \vee \cdots \vee l_{rk_r})$$

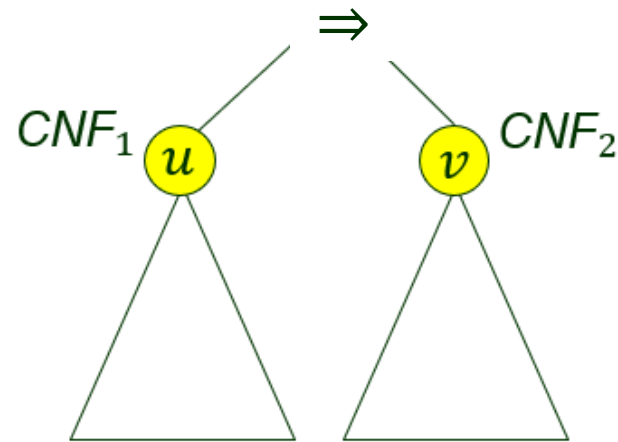


$$\neg CNF \equiv \bigwedge_{\substack{1 \leq j_1 \leq k_1 \\ \vdots \\ 1 \leq j_r \leq k_r}} (\neg l_{1j_1} \vee \neg l_{2j_2} \vee \cdots \vee \neg l_{rj_r})$$

- If l_{ij} is a negative literal, i.e., $l_{ij} = \neg p_{ij}$, then $\neg l_{ij}$ reduces to p_{ij} because the two occurrences of $\neg$ cancel out.

Case 4

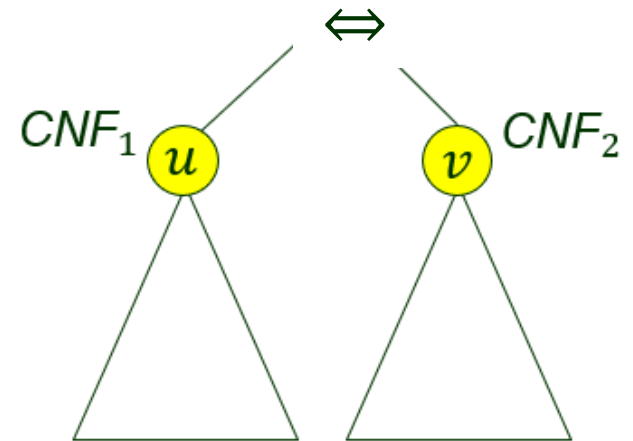
The node n stores $\Rightarrow$.



- Logically equivalent to $\neg CNF_1 \vee CNF_2$.
- First, convert $\neg CNF_1$ into conjunctive normal form CNF'_1 (see case 3).
- Then, convert the disjunction $CNF'_1 \vee CNF_2$ into conjunctive normal form (see case 2)

Case 5

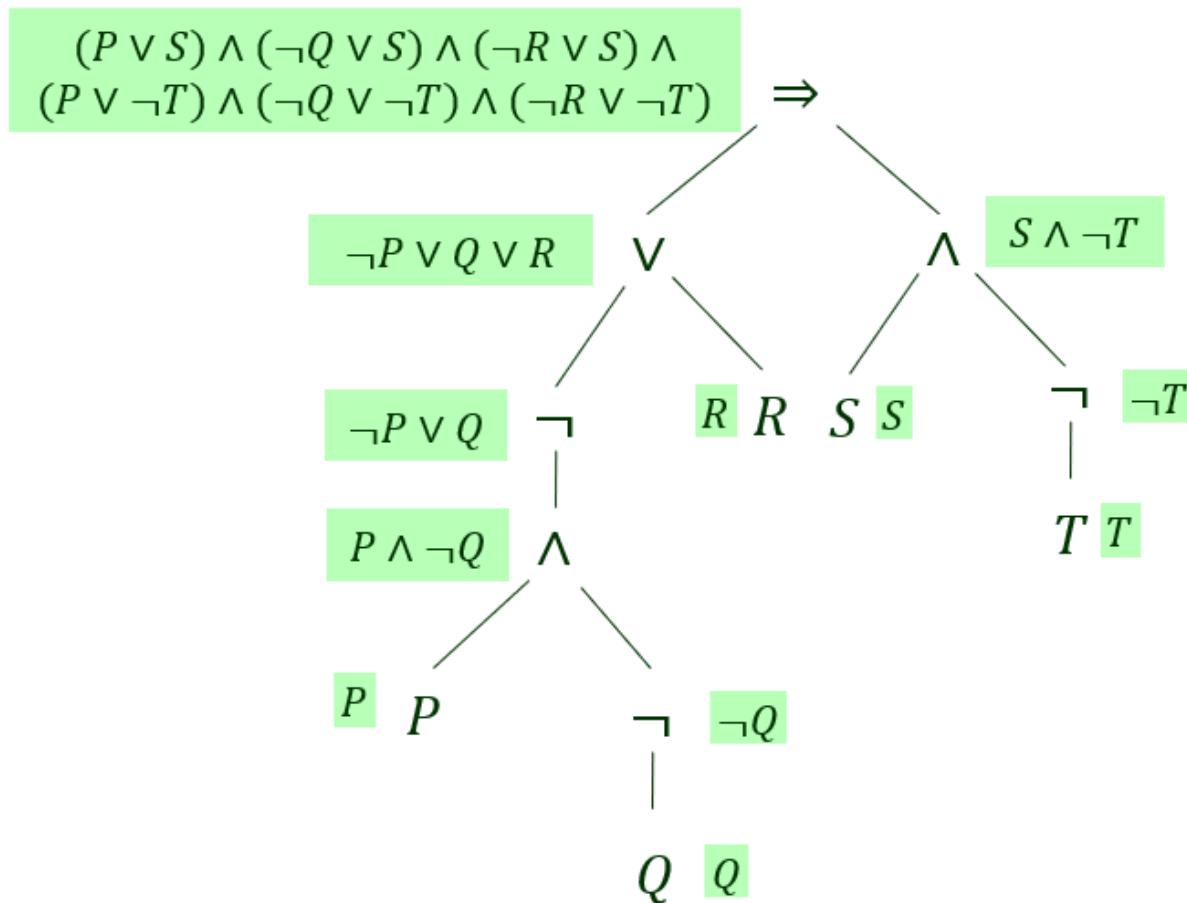
The node n stores $\Leftrightarrow$.



- Logically equivalent to $(CNF_1 \Rightarrow CNF_2) \wedge (CNF_2 \Rightarrow CNF_1)$.
- Convert $CNF_1 \Rightarrow CNF_2$ and $CNF_2 \Rightarrow CNF_1$ separately into conjunctive normal forms CNF'_1 and CNF'_2 (see case 4).
- Return $CNF'_1 \wedge CNF'_2$.

Example

$\neg(P \wedge \neg Q) \vee R \Rightarrow S \wedge \neg T$ has the following conjunctive normal form:



III. Proof by Resolution – An Example

KB:

P
 $P \rightarrow (Q \vee R)$
 $Q \rightarrow S$
 $R \rightarrow (S \wedge T)$

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Q: $KB \vdash S$?

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Q: $KB \vdash S$?

1. Converting sentences to CNF

III. Proof by Resolution – An Example

KB:

$$\begin{array}{l} P \\ P \rightarrow (Q \vee R) \\ Q \rightarrow S \\ R \rightarrow (S \wedge T) \end{array} \quad \text{-----} \rightarrow \quad \neg P \vee Q \vee R$$

Q: $KB \vdash S$?

1. Converting sentences to CNF

III. Proof by Resolution – An Example

KB:

P

$P \rightarrow (Q \vee R) \quad \text{-----} \rightarrow \neg P \vee Q \vee R$

$Q \rightarrow S \quad \text{-----} \rightarrow \neg Q \vee S$

$R \rightarrow (S \wedge T)$

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Q: $KB \vdash S$?

1. Converting sentences to CNF

KB: P

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$(\neg R \vee S) \wedge (\neg R \vee T)$

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Q: $KB \vdash S$?

1. Converting sentences to CNF
2. Spilt each conjunction into clauses.

KB: P

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$\neg Q \vee S$

$(\neg R \vee S) \wedge (\neg R \vee T)$

III. Proof by Resolution – An Example

KB:

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Q: $KB \vdash S$?

1. Converting sentences to CNF 2. Spilt each conjunction into clauses.

KB:

P

$\neg P \vee Q \vee R$

$\neg Q \vee S$

$(\neg R \vee S) \wedge (\neg R \vee T) \quad \left\{ \begin{array}{l} \neg R \vee S \\ \neg R \vee T \end{array} \right.$

III. Proof by Resolution – An Example

KB:

P

$P \rightarrow (Q \vee R) \quad \text{-----} \rightarrow \neg P \vee Q \vee R$

$Q \rightarrow S \quad \text{-----} \rightarrow \neg Q \vee S$

$R \rightarrow (S \wedge T) \quad \text{-----} \rightarrow \neg R \vee (S \wedge T)$

$\text{-----} \rightarrow (\neg R \vee S) \wedge (\neg R \vee T)$

Q: $KB \vdash S$?

1. Converting sentences to CNF
2. Split each conjunction into clauses.

KB: P

$\neg P \vee Q \vee R$

$\neg Q \vee S$

$(\neg R \vee S) \wedge (\neg R \vee T)$

KB: P

$\neg P \vee Q \vee R$

$\neg Q \vee S$

$\neg R \vee S$

$\neg R \vee T$

Proof by Resolution

KB (updated):

- (1) P
- (2) $\neg P \vee Q \vee R$
- (3) $\neg Q \vee S$
- (4) $\neg R \vee S$
- (5) $\neg R \vee T$

(1) P

(2) $\neg P \vee Q \vee R$

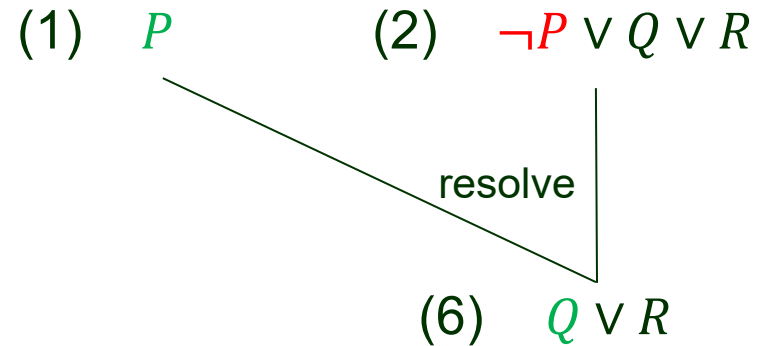
Q: $KB \vdash S$?

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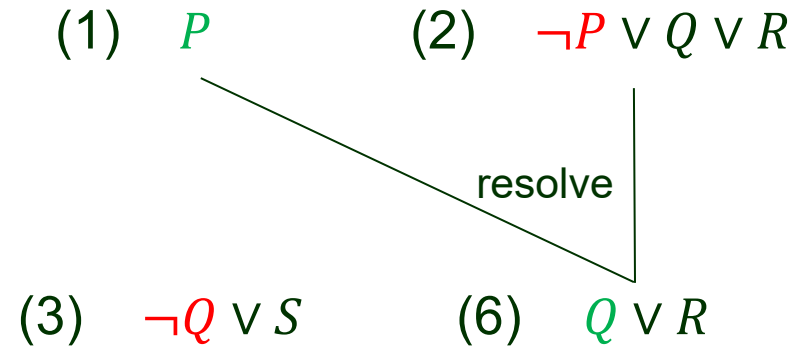


Proof by Resolution

KB (updated):

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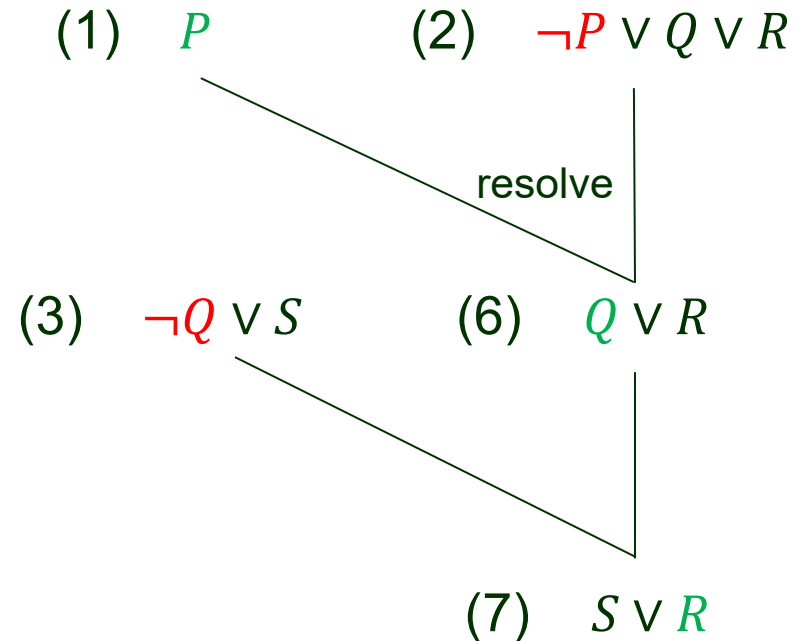


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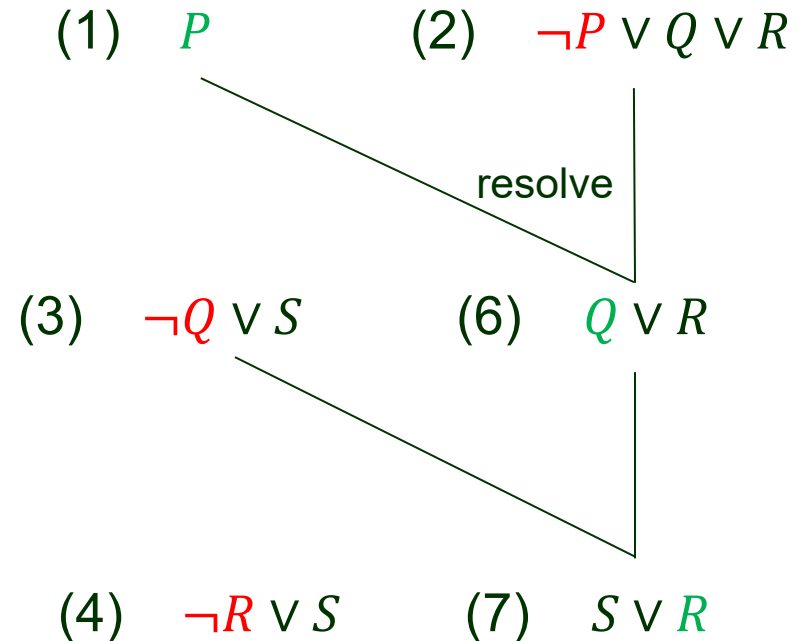


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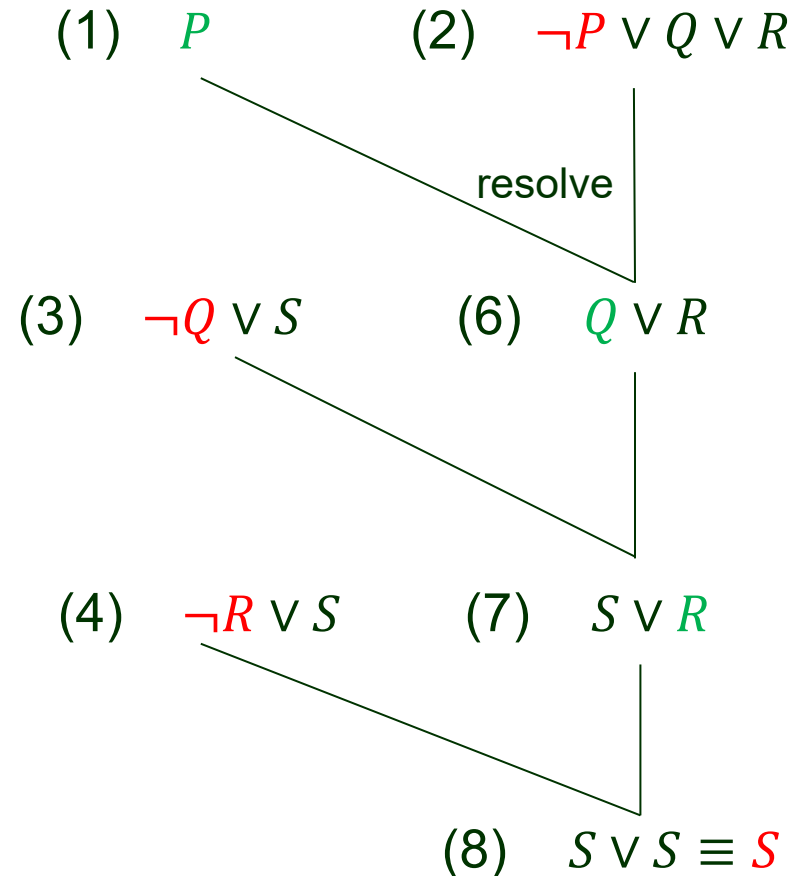


Proof by Resolution

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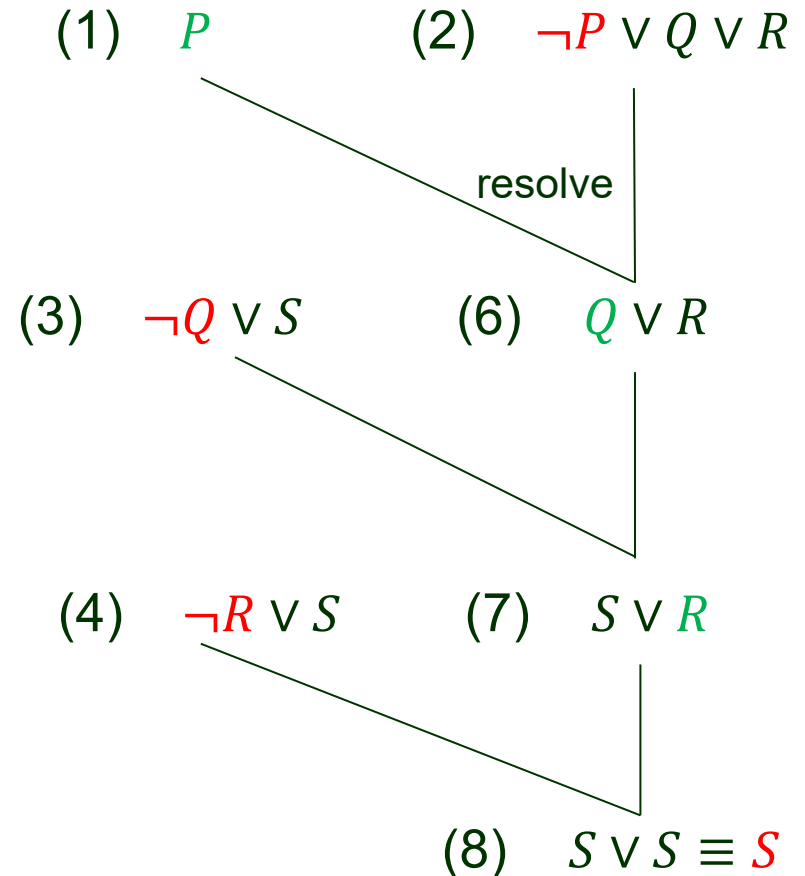


Proof by Resolution

KB (updated):

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$Q: KB \vdash S ?$



Resolution tree

Resolution Refutation

(Proof by contradiction)

To show that $KB \models \alpha$, we show that $KB \wedge \neg\alpha$ is unsatisfiable. .

Resolution Refutation

(Proof by contradiction)

To show that $KB \models \alpha$, we show that $KB \wedge \neg\alpha$ is unsatisfiable. .

KB (about a summer day):

- (1) If it is raining and you are outside then you will get wet.
- (2) If it is warm and there is no rain then it is a pleasant day.
- (3) You are not wet.
- (4) You are outside.
- (5) It is a warm day.

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(Proof by contradiction)

To show that $KB \models \alpha$, we show that $KB \wedge \neg\alpha$ is unsatisfiable. .

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Prove

It is a pleasant day.

KB in Propositional Sentences

KB (rewritten):

- (1) $(\text{rain} \wedge \text{outside}) \Rightarrow \text{wet}$
- (2) $(\text{warm} \wedge \neg \text{rain}) \Rightarrow \text{pleasant}$
- (3) $\neg \text{wet}$
- (4) outside
- (5) warm

KB in Propositional Sentences

KB (rewritten):

```
(1) ( rain  $\wedge$  outside )  $\Rightarrow$  wet  
(2) ( warm  $\wedge$   $\neg$ rain )  $\Rightarrow$  pleasant  
(3)  $\neg$ wet  
(4) outside  
(5) warm
```



converted into clauses

```
(1)  $\neg$ rain  $\vee$   $\neg$ outside  $\vee$  wet  
(2)  $\neg$ warm  $\vee$  rain  $\vee$  pleasant  
(3)  $\neg$ wet  
(4) outside  
(5) warm
```

KB in Propositional Sentences

KB (rewritten):

```
(1) ( rain  $\wedge$  outside )  $\Rightarrow$  wet  
(2) ( warm  $\wedge$   $\neg$ rain )  $\Rightarrow$  pleasant  
(3)  $\neg$ wet  
(4) outside  
(5) warm
```



converted into clauses

```
(1)  $\neg$ rain  $\vee$   $\neg$ outside  $\vee$  wet  
(2)  $\neg$ warm  $\vee$  rain  $\vee$  pleasant  
(3)  $\neg$ wet  
(4) outside  
(5) warm
```

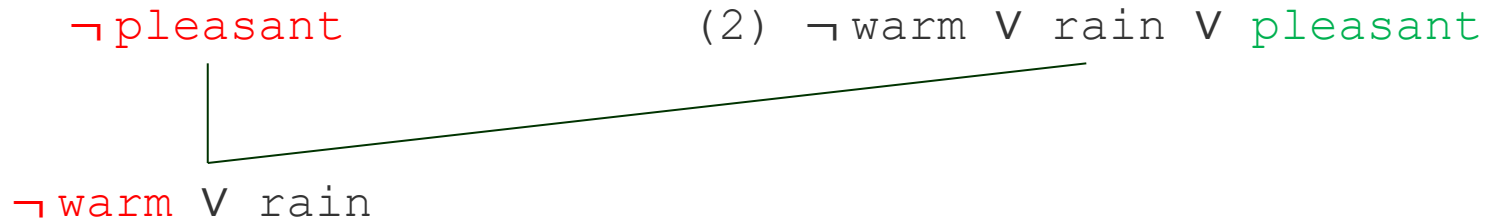
We add $\neg$ pleasant to KB and try to derive an empty clause $\emptyset$.

Resolution Refutation Tree

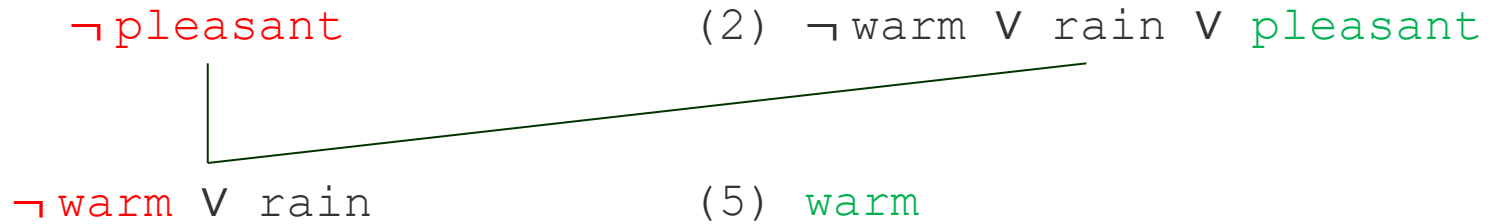
$\neg$ pleasant

(2) $\neg$ warm $\vee$ rain $\vee$ pleasant

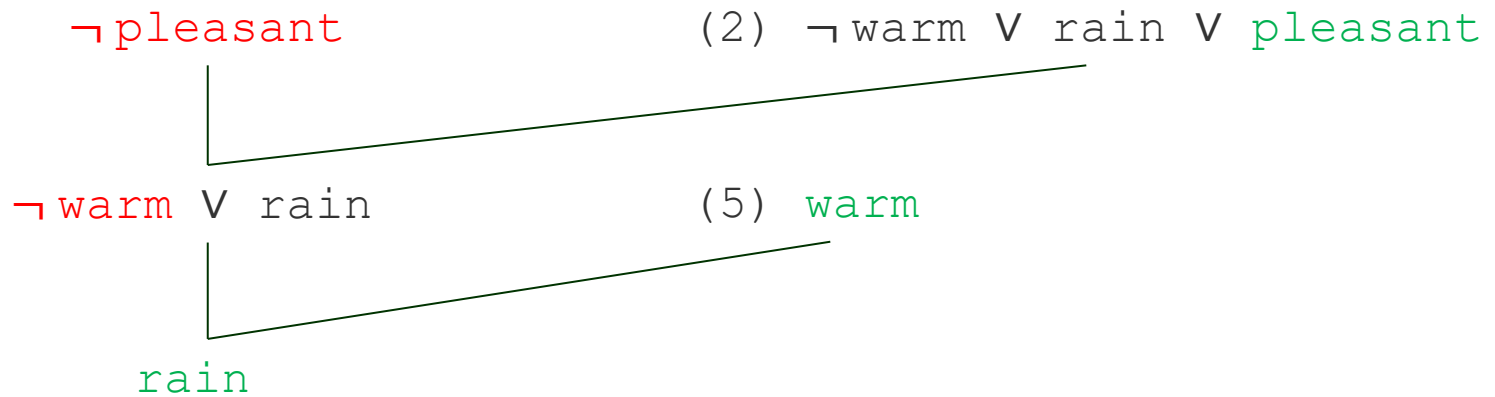
Resolution Refutation Tree



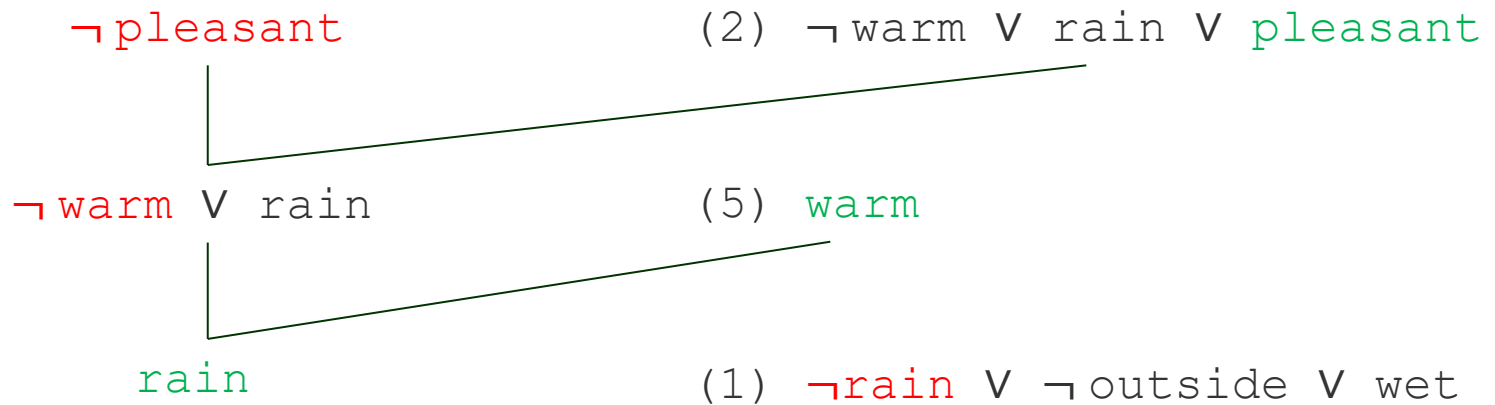
Resolution Refutation Tree



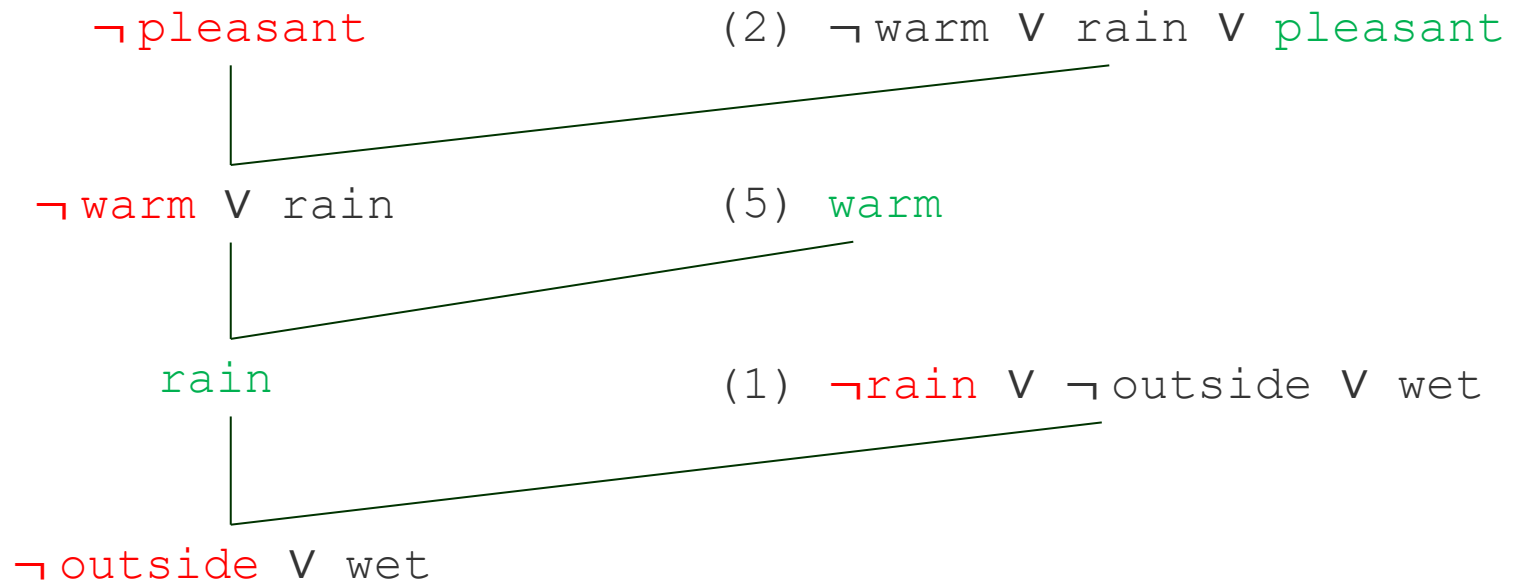
Resolution Refutation Tree



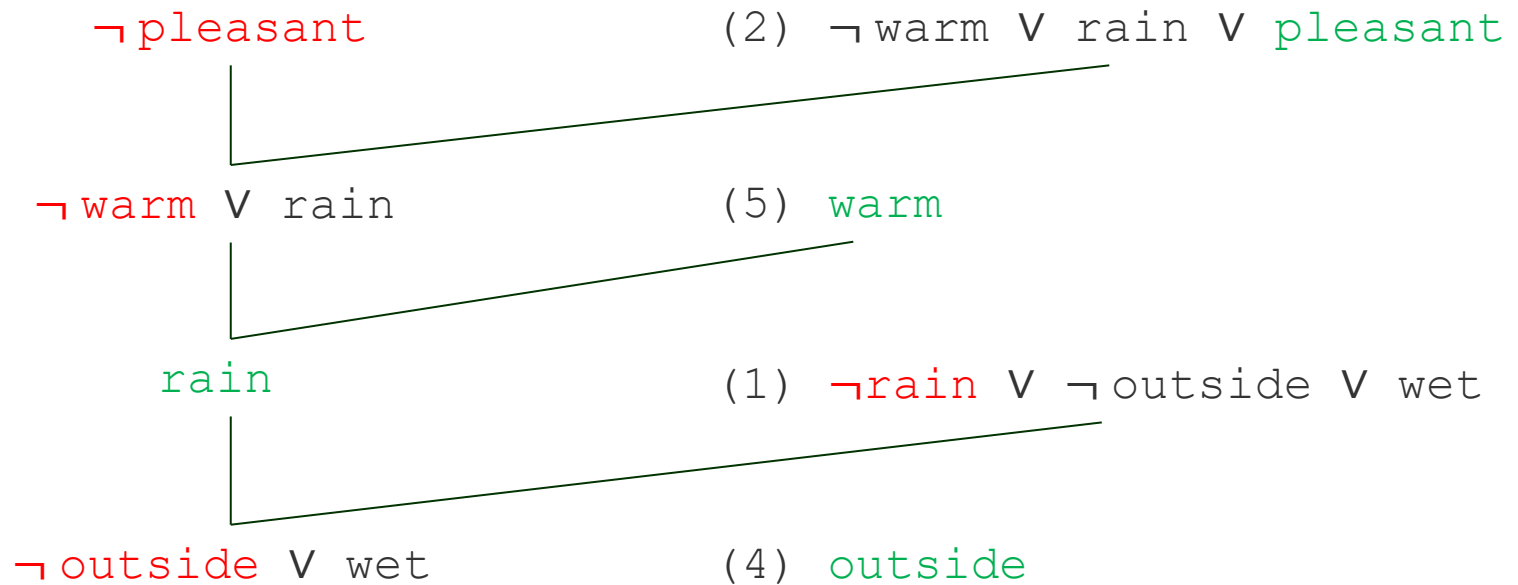
Resolution Refutation Tree



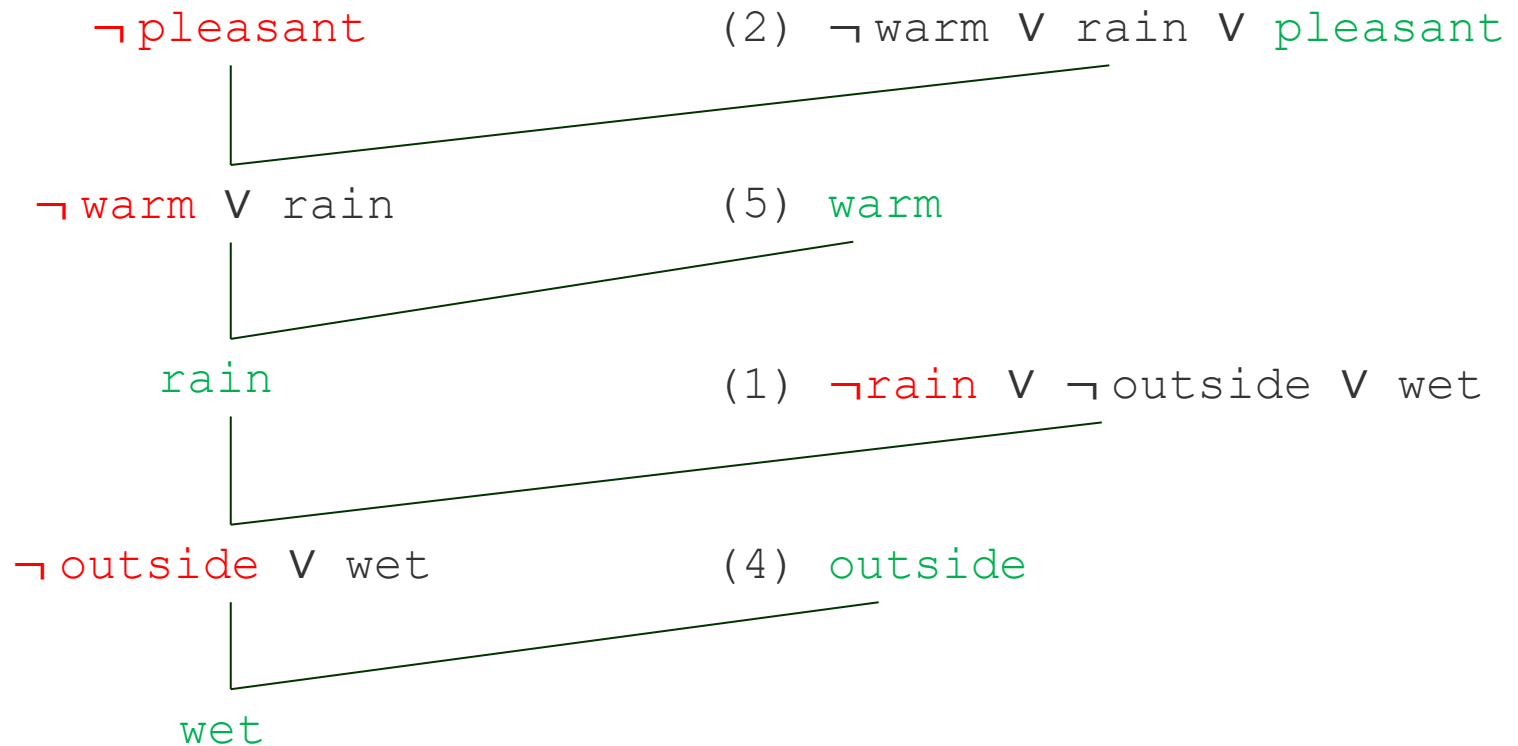
Resolution Refutation Tree



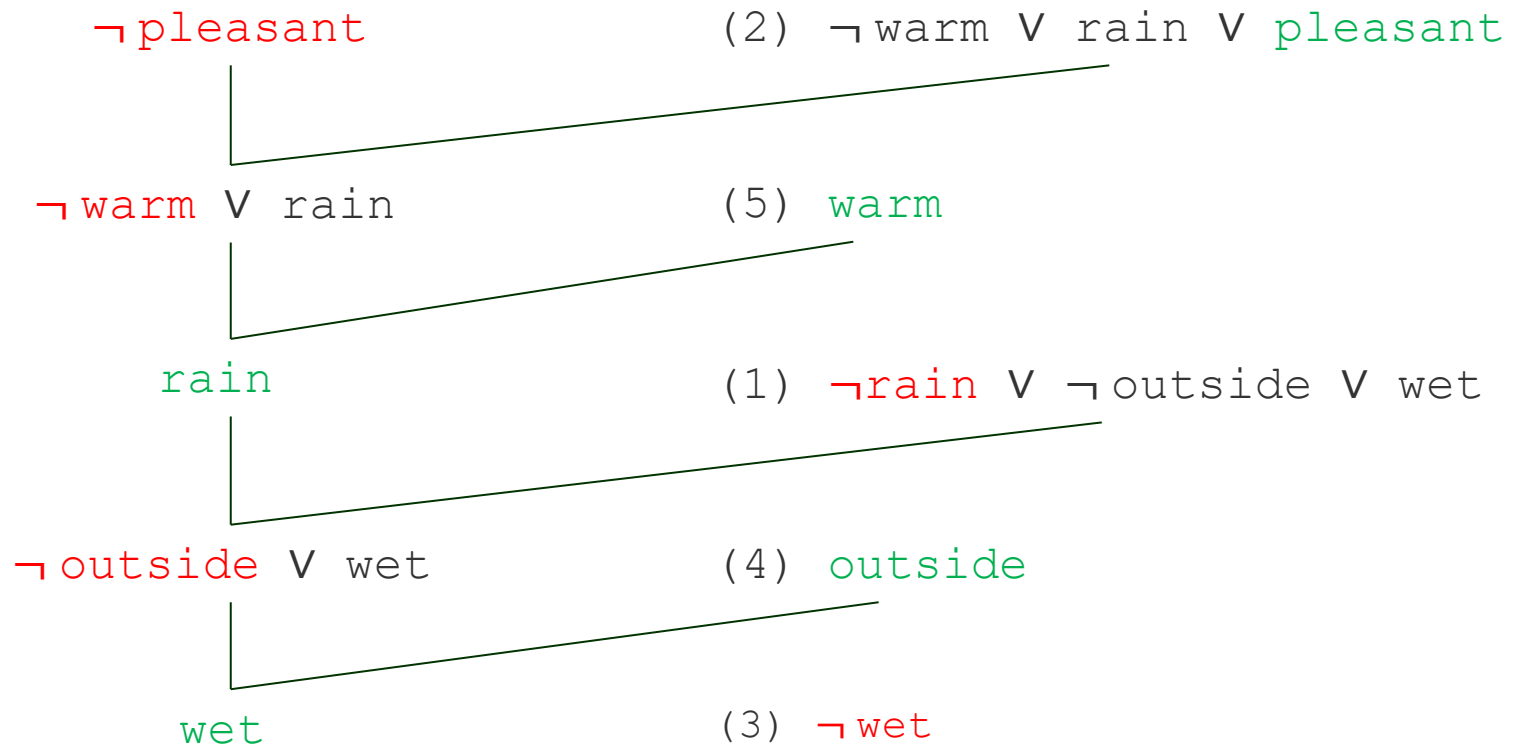
Resolution Refutation Tree



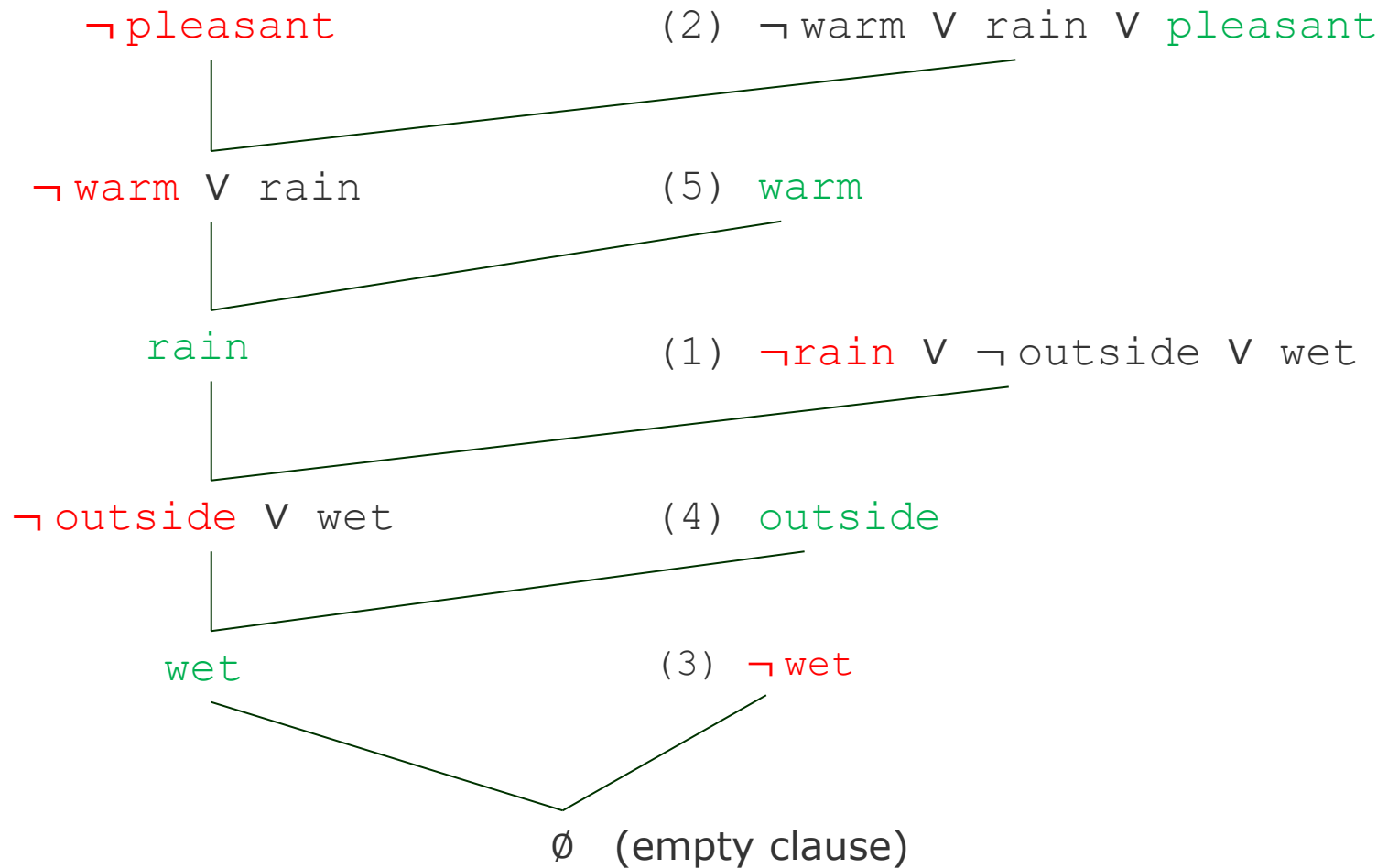
Resolution Refutation Tree



Resolution Refutation Tree



Resolution Refutation Tree



Resolution Refutation Tree

